1) $\underset{8}{\_}\ \underset{8}{\_}\ \underset{7}{\_}\ \underset{5}{\_}$

$8 \cdot 8 \cdot 7 \cdot 5 = 64 \cdot 35 \boxed{= 2240}$ ✓

$$\begin{array}{r} 64 \cdot \\ 35 \\ \hline 320 \\ 192 \\ \hline 2240 \end{array}$$

2) $\underline{1}\ \underline{1}\ \_\ \_\ \_\ \_\ \_\ \_\ \_\ \underline{0}\ \underline{0}\ \underline{0}$ a) $2^5 = 32$ ✓

b) $\underset{1}{\_}\ \_\ \_\ \_\ \_\ \_\ \_\ \_\ \_\ \_\ \underset{2}{\_} = 2^9 = 512$ ✓

2^9

c) $\overbrace{\_\ \_\ \_\ \_\ \_\ \_\ \_\ \_\ \_}^{2^9}\ \_\ \_$ 2^9 ways - if nu odd => 1 else => 0

=> 512

d) 512 as well $\overline{\_\ \_\ \_}$
?

0 de 1 - 1
2 de 1 - $P_3 = 3! = 6$ | => 7 nos

e) $\underset{2}{\_}\ \underset{1}{\_}\ \underset{1}{\_}\ \underset{1}{\_} = 2 \cdot 1 \cdot 1 \cdot 1 = 2$ -> 2 string ✓

3) 8 7 albe 1 albastră

1, 3, 5, 7

$C_8^1 = 8$ moduri

$C_8^3 = \frac{8!}{3!5!} = \frac{\cancel{6} \cdot 7 \cdot \cancel{8}}{\cancel{2} \cdot \cancel{3}} = 56$

$C_8^5 = \frac{8!}{5!3!} = 56$

$C_8^7 = 8$ moduri

=> $112 + 16 = 128$ ✓

4) $\{0, 1, 2, 3, 4, 5, 6, 7\}$

$$\underset{3}{\_\_}\ \underset{4}{\_\_}\ \underset{3}{\_\_}\ \underset{3}{\_\_}\ \underset{2}{\_\_} = 3^3 \cdot 2^3 = 216$$

$$\underset{4}{\_\_}\ \underset{4}{\_\_}\ \underset{3}{\_\_}\ \underset{3}{\_\_}\ \underset{2}{\_\_} = 3^2 \cdot 2^5 = 9 \cdot 32 = 288$$

$\Rightarrow$ 504 moduri

5) $A = \{1,2,3\}$ a) $f: A \to B$

$B = \{1,2,3,4\}$ $4^3 = 64$

c) $g: B \to A$

$3^4 = 81$

b) $1 \to 4$, $2 \to 3$, $3 \to 2$ $\Rightarrow 4 \cdot 3 \cdot 2 = 24$

d) Since $|A| < |B| \Rightarrow$ at least one element from B will have the same image with another one.

$\Rightarrow$ g is not injective

Also, f is not surjective because at least one element won't be the image of an element of A.

e) $1 \to$, $2 \to$, $3 \to 2$ $\Rightarrow$ 6 functions

$$f(1) < \underset{|}{f(2)} < \underset{\downarrow\\ 2}{f(3)}$$

$C_4^3 = 4$

6) $\{4,5,6,7,8,9,10\}$

a) $2^7 = 128 \underset{\{\emptyset\}}{\underset{\downarrow}{-1}} \underset{\{3\}}{\underset{\downarrow}{+1}} = 128$

b) $\{1,2,\ldots,10\} = \{$?

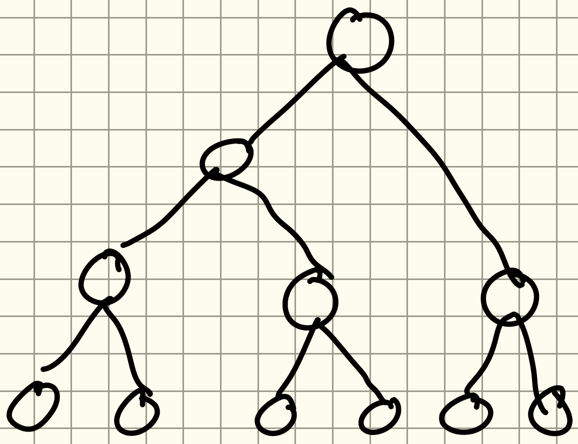
1 -> 9 opțiuni   3 -> 5 opțiuni   5 -> 1 opțiune
2 -> 7 opțiuni
4 -> 3 opțiuni   => Total $= 9 \cdot 7 \cdot 5 \cdot 3$
$= 63 \cdot 15$
$= 945$

7) a) 64 -> 32 meciuri
32 -> 16 meciuri
16 -> 8 meciuri
8 -> 4 meciuri
4 -> 2 meciuri
2 -> 1 meci

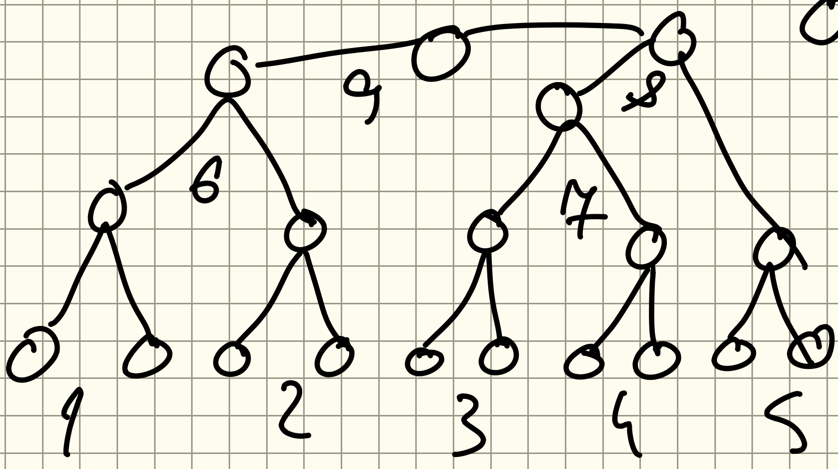
$1+2+\ldots+32$

$1+2+\ldots+2^5$

$1 \cdot \frac{2^6-1}{2-1} = 63$



b) 100 -> 50 de meciuri
50 -> 25 de meciuri
25 -> 12 meciuri



c) n - 1 meciuri

9

20

8)

3 Cei   3, 2, 1

_ _ _
↓ ↓

3
4000
3100
2110
2200
2111
1111

? ? ?

n:

2 0
— —
1 1
— —

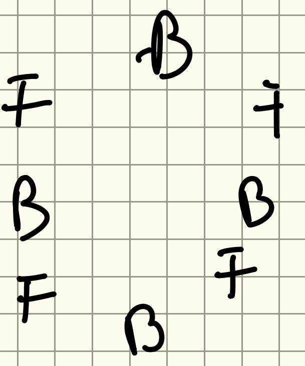
$\overline{3}\ \overline{0}\ \overline{0}$
2 1 0
1 1 1

9)

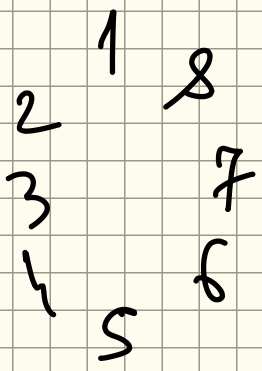
B F B F B F B F

F B F B F B F B

? ? ? ?



2



10) $M = \{1, 2, \dots, n\}$

$M = \{1, 2, 3\}$

$A = \{1, \}$

B